

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 38

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Renewable: re-forms naturally relatively quickly so use does not necessarily reduce future availability (solar, wind, tidal, geothermal, biofuel) (1) • Depletable: use can reduce future availability - all non-renewables (fossil fuels, uranium) plus renewables exploited unsustainably (1) • Wood / biofuels are renewable but depletable if forests are felled faster than they regrow (1)	3
01.2	<i>1 mark for each point.</i> • Other factors (technology, location, cost) restrict how much can be harnessed (1) • Examples: coal too deep to mine; winds high above the ground / far offshore where turbines cannot be placed; waves in deep water where equipment cannot be anchored; low-intensity sunlight that cannot produce high temperatures (1) • New technologies or higher prices may make these usable in future (1)	3
01.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Wind: strong reliable winds - shallow seas, open plains, uplands; low land-use conflict (not near urban areas, ecologically sensitive or scenic areas); near the grid (1) • HEP: high reliable rainfall, large catchment, narrow valley for a small dam with a large basin, impermeable stable bedrock (1) • Solar: high light intensity, low cloud cover (1) • Wave: coasts with strong reliable prevailing winds and a long fetch (1) • Tidal: large tidal range, coastal features (estuaries, channels) that focus the flow (1) • Geothermal: hot rocks near the surface / recent volcanic activity (1) • Biofuels: nearby farmland / forest / urban waste to avoid transporting bulky low-density fuel (1)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-177; AQA spec 3.3.2.	
02.1	<i>1 mark for each point.</i> - Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended (1) - Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges (1) - The particles clump together (flocs) and settle in the clarifier tank (1)	3
02.2	<i>1 mark for each point.</i> - Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources (1) - Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed; improves taste (1)	2
02.3	<i>1 mark for each point.</i> - Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) (1) - Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution) (1)	2
02.4	<i>1 mark for each point.</i> - Ion exchange: toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) (1) - Reverse osmosis / distillation (desalination) - expensive, energy-intensive (1) - Activated carbon: organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	
03.1	<i>1 mark for each point.</i> - Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier (1) - Standardise the noise source: same cars / same race or a standard test noise; same time of day, wind speed / direction, weather; same height of meter above ground (1) - Take repeated readings at each distance and calculate means (and standard deviation) to assess variability (1) - Use a measure such as LA10 or Leq over a set period rather than a single peak reading (1) - Compare reduction in dB relative to the control at each distance; plot noise level against distance for each barrier; statistical test (t-test) to assess significance (1)	5
03.2	<i>1 mark.</i> Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier (1)	1
03.3	<i>1 mark for each point.</i> - LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods (1) - LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level (1)	2

Question	Marking guidance	Marks
03.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Double / triple glazing (1) • Acoustic insulation in walls / roof (1) • Bedrooms on the side facing away from the track (1) • Dense hedges / trees and earth mounds in gardens (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.	
04.1	<i>1 mark for each point.</i> • Net held with its straight edge on the river bed facing the current (open to the flow) (1) • The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) (1) • Same substrate type, water depth and flow rate at both sites; same net mesh; repeat samples; invertebrates identified with a key and counted (1)	3
04.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Difficult to standardise the vigour / area kicked (1) • Buried organisms or those fixed to stones are not dislodged (1) • Some organisms swim and avoid the net (1) • Flow rate varies and affects capture (1) • Fragile organisms destroyed (1)	3
04.3	<i>1 mark for each point.</i> • A fixed area of bed is enclosed by a frame so the sampled area is standardised; side nets reduce escapes (1) • But only works in fast enough flow and is hard to set on rough substrates / deep water (1)	2
04.4	<i>1 mark for each point.</i> • Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event (1) • Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
05.1	<i>1 mark for each point.</i> • Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) (1) • To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites (1)	2
05.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Sea current velocity and direction (1) • Properties of the radionuclides: half-life, solubility, tendency to adsorb onto sediments / bioaccumulate (1) • Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people (1) • Geology / hydrology of the coast; sediment deposition areas (mud flats) (1) • Human activities in the area: fishing, seaweed collection, beach use (1)	3

Question	Marking guidance	Marks
05.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) (1) - Monitor their exposure / the concentration of radioactive material in the foods they eat or their body (1) - If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk; if not, emissions are controlled (1) - CGM assesses exposure before health effects occur (not used to detect damage) (1)	3
05.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Seawater (1) - Sediment / mud (1) - Seaweed (1) - Shellfish / molluscs / fish (1) - Milk, vegetables or meat from nearby farms (1) - Atmospheric dust (1) - Soil (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	
06.1	<i>1 mark for each point.</i> - CO₂ rises from about 412 ppm in January to a peak of 416 ppm in May, falls to a minimum of 407 ppm in September, then rises again (1) - During spring / summer, photosynthesis by plants (growing season) removes CO₂ faster than respiration releases it, so the concentration falls (1) - In autumn / winter photosynthesis is reduced (leaf fall) but respiration and decomposition continue, so CO₂ rises (1) - The atmosphere is in dynamic equilibrium: the concentration fluctuates around a mean; more land / vegetation in the northern hemisphere makes the cycle large (1)	4
06.2	<i>1 mark for each point.</i> - Natural processes that add gases (respiration, combustion, volcanoes, denitrification) are balanced by processes that remove them (photosynthesis, dissolving, nitrogen fixation) (1) - So the average composition stays the same over long timescales although the rates vary and concentrations fluctuate around a mean (1)	2
06.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Absorbs harmful UV / solar wind radiation in the stratosphere (ozone layer) (1) - Greenhouse gases delay the escape of infrared, keeping the surface warm (33 °C warmer than without) (1) - Atmospheric pressure allows liquid water to exist (1) - Distributes heat from the tropics to higher latitudes by winds (1) - Transports water vapour to areas that would otherwise be dry (1)	2
06.4	<i>1 mark for each point.</i> - Troposphere (1) - Stratosphere (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89; AQA spec 3.2.1.1.	

Question	Marking guidance	Marks
07.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(2900 - 1400) / 1400 \times 100$ (1) • = 107% (accept 107.1) (1)	2
07.2	<i>1 mark for each point.</i> • Large decrease from 24 per year (1975-79) to 2 per year (2010-14) (1) • Steepest fall between the late 1970s and 1980s; slower decline / levelling off after 1995 (1)	2
07.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Double-hulled tankers required for all large tankers (1) • Improved navigation (GPS, AIS) and offshore shipping routes (1) • Inert gas systems reducing explosions (1) • Twin engines / rudders reducing groundings after mechanical failure (1) • Stricter regulation, inspection and polluter-pays liability (eg after Exxon Valdez) (1) • Separate ballast tanks / crude oil washing (1)	3
07.4	<i>1 mark for each point.</i> • Most oil pollution comes from dispersed / diffuse sources not tanker accidents: waste lubricating oil, industrial oil in drains, road runoff (1) • Routine operations: small spills, pipeline leaks, storage tank leakage, oil rig accidents (eg Deepwater Horizon) (1) • Growth in oil use and transport means more small releases even if large accidents are rarer (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-262; AQA spec 3.4.3.2.7.	
08.1	<i>1 mark.</i> • The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) (1)	1
08.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions (1) • Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition (1) • Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils (1) • pH / presence of other chemicals may increase or decrease the rate of degradation (1) • Adsorption onto clay / organic matter may protect from degradation and increase persistence (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
09.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Public opposition following accidents (Three Mile Island 1979, Chernobyl 1986, Fukushima 2011) (1) • Very complex, expensive technology needing an advanced industrial infrastructure (1) • Concerns about links between civil nuclear materials and weapons / terrorism (1) • Uncertainty over permanent disposal of radioactive waste (1) • Uncertain decommissioning costs (1)	3

Question	Marking guidance	Marks
09.2	<i>1 mark for each point.</i> • Only 0.7% of natural uranium is fissile U-235; the remaining 99.3% is non-fissile U-238 (a 'fertile' fuel) (1) • Neutron bombardment in the reactor converts U-238 to fissile plutonium-239 (via beta decays) (1) • So much more energy is harnessed from the mined uranium; fast reactors need no moderator but are complex and costly (1)	3
09.3	<i>2 marks for each point (partial credit for a partly correct point).</i> • Advantages: thorium is three times more abundant; much less radioactive waste with shorter half-lives; harder to make weapons material; no fuel enrichment needed (2) • Disadvantages: thorium-232 is not fissile so must be bred into uranium-233 (slow breeding rate); U-233 is an alpha emitter so very hazardous to handle; less developed technology with high development costs (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 192-194; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Control of greenhouse gases: energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • Geoengineering: white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • International agreements: Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation; carbon trading and taxes; comparison of per capita carbon footprints between countries • Evidence of limited success: global CO₂ concentrations still rising (about 280 ppm pre-industrial to over 410 ppm); temperatures up about 1 °C; emissions in some developed countries falling but rising in developing economies • Reasons for difficulty: fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • Judgement: partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise; tipping points make urgency greater	9
10.2	<i>1 mark.</i> • The total greenhouse gas / CO₂ emissions caused by an individual, organisation, product or country (often expressed per capita per year) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
11.1	Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Composition and structure: N₂, O₂, Ar, CO₂, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry • Energy processes: UV absorbed by O / O₂ / O₃ dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo • Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements • Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • Other threats: aerosols, smoke, land-use change; interconnection of processes • Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	25

Question	Marking guidance	Marks
11.2	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25